ELSEWEDY
UNIVERSITY
OF TECHNOLOGY
- POLYTECHNIC OF EGYPT -

GEN211 Probability and Statistics

Lecture 7: The Normal Distribution and Applications

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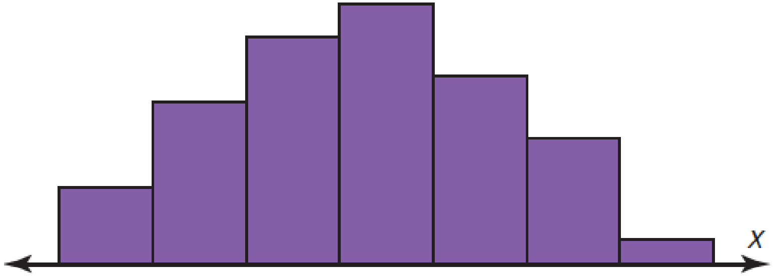
Lecture contents

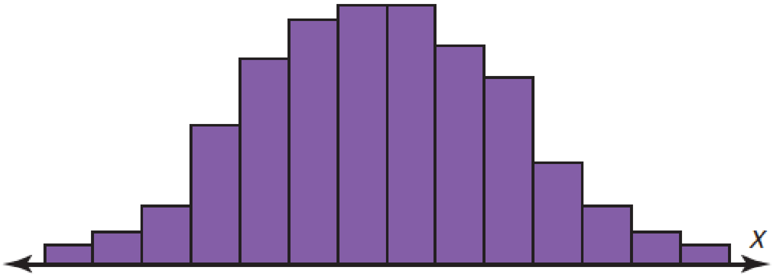
- The normal distribution
- The central limit theorem

The Normal Distribution

Introduction

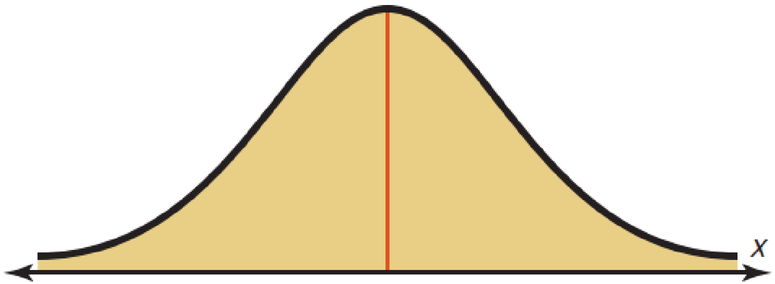


(a) Random sample of 100 women



(b) Sample size increased and class width decreased

(c) Sample size increased and class width decreased further



(d) Normal distribution for the population

Histograms and Normal Model for the Distribution of Heights of Adult Women

Introduction, revisit lecture 1, slide 14

Recall that a discrete variable cannot assume all values between any two given values of the variables. On the other hand, a continuous variable can assume all values between any two given values of the variables. Examples of continuous variables are the height of adult men, body temperature of rats, and cholesterol level of adults. Many continuous variables, such as the examples just mentioned, have distributions that are bell-shaped, and these are called *approximately normally distributed variables*. For example, if a researcher selects a random sample of 100 adult women, measures their heights, and constructs a histogram, the researcher gets a graph similar to the one shown in Figure (a). Now, if the researcher increases the sample size and decreases the width of the classes, the histograms will look like the ones shown in Figure (b) and (c). Finally, if it were possible to measure exactly the heights of all adult females in the United States and plot them, the histogram would approach what is called a *normal distribution curve*, as shown in Figure (d). This distribution is also known as a *bell curve* or a *Gaussian distribution curve*, named for the German mathematician Carl Friedrich Gauss (1777–1855), who derived its equation.

Normal distributions

If a random variable has a probability distribution whose graph is continuous, bell-shaped, and symmetric, it is called a **normal distribution.** The graph is called a *normal distribution curve*.

The mathematical equation for a normal distribution is

$$y = \frac{e^{-(X-\mu)^2/(2\sigma^2)}}{\sigma\sqrt{2\pi}}$$

where $e \approx 2.718$ ($\approx$ means "is approximately equal to")

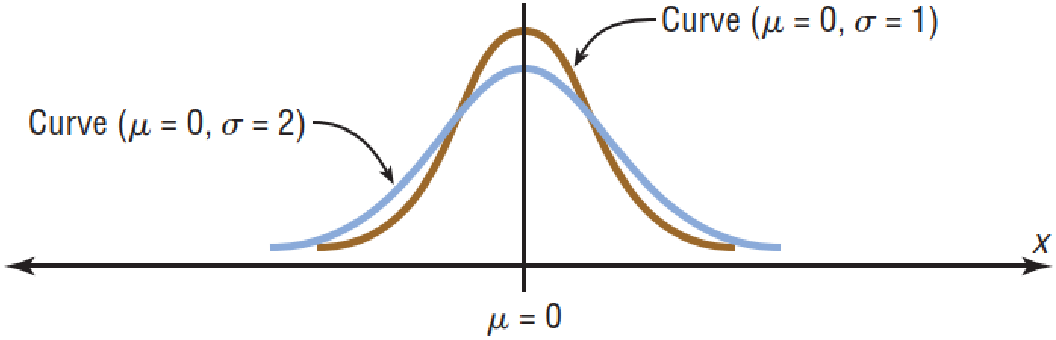
$\pi \approx 3.14$

μ = population mean

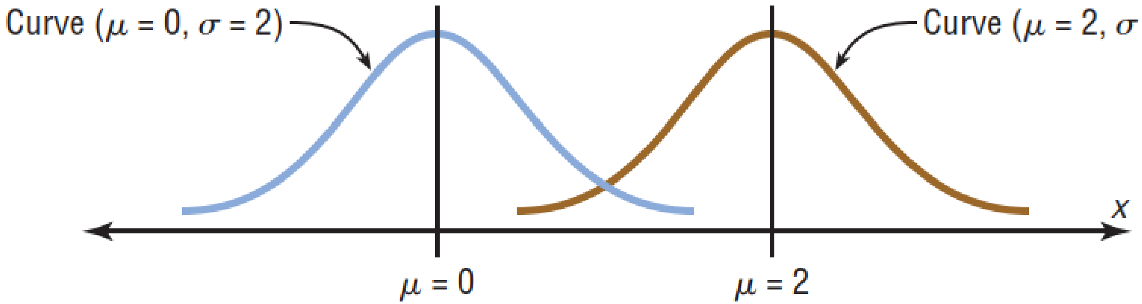
σ = population standard deviation

The shape and position of a normal distribution curve depend on two parameters, the *mean* and the *standard deviation.* Each normally distributed variable has its own normal distribution curve, which depends on the values of the variable's mean and standard deviation.

Shapes of normal distributions



(a) Same means but different standard deviations



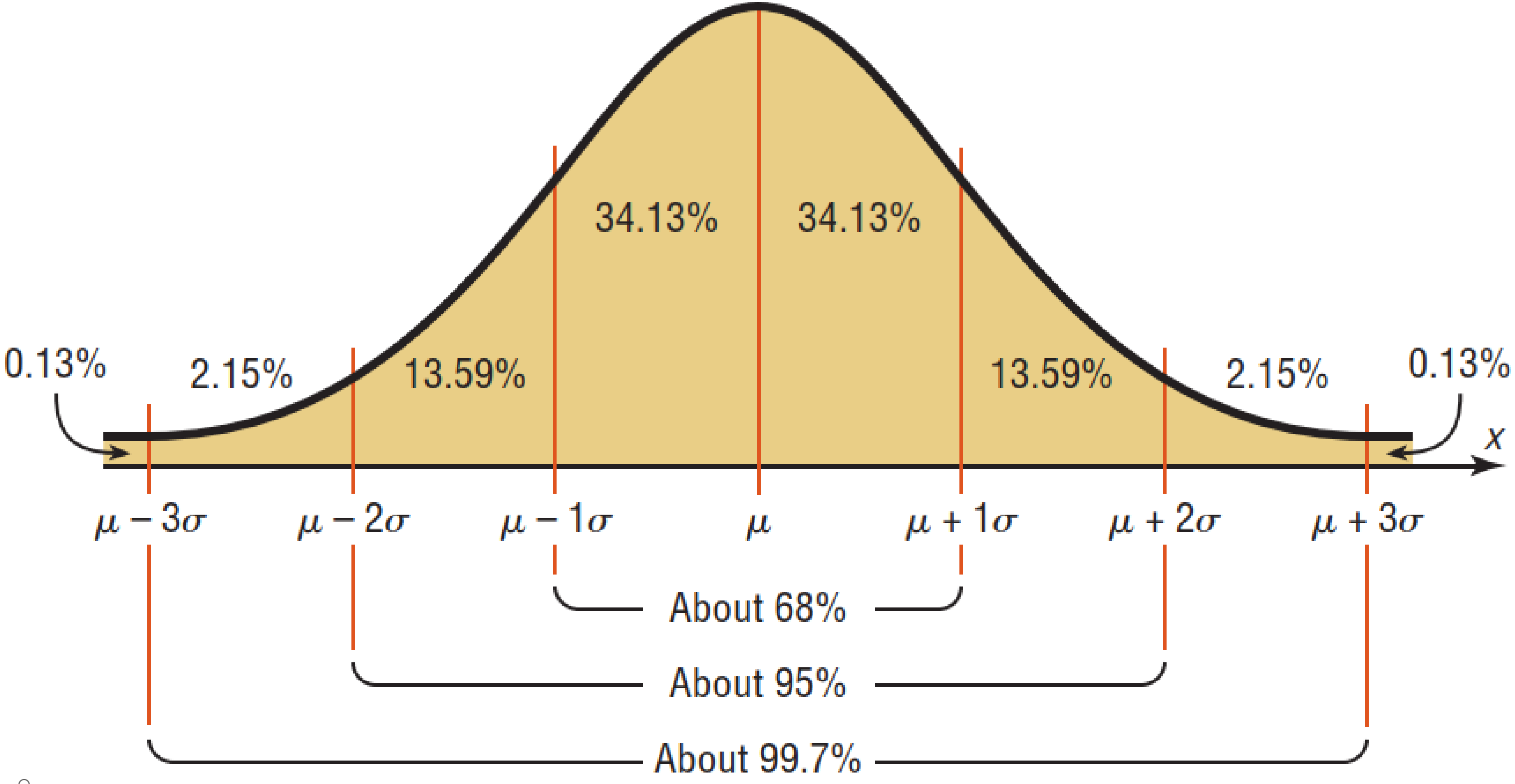
(b) Different means but same standard deviations

Properties of the theoretical normal distribution

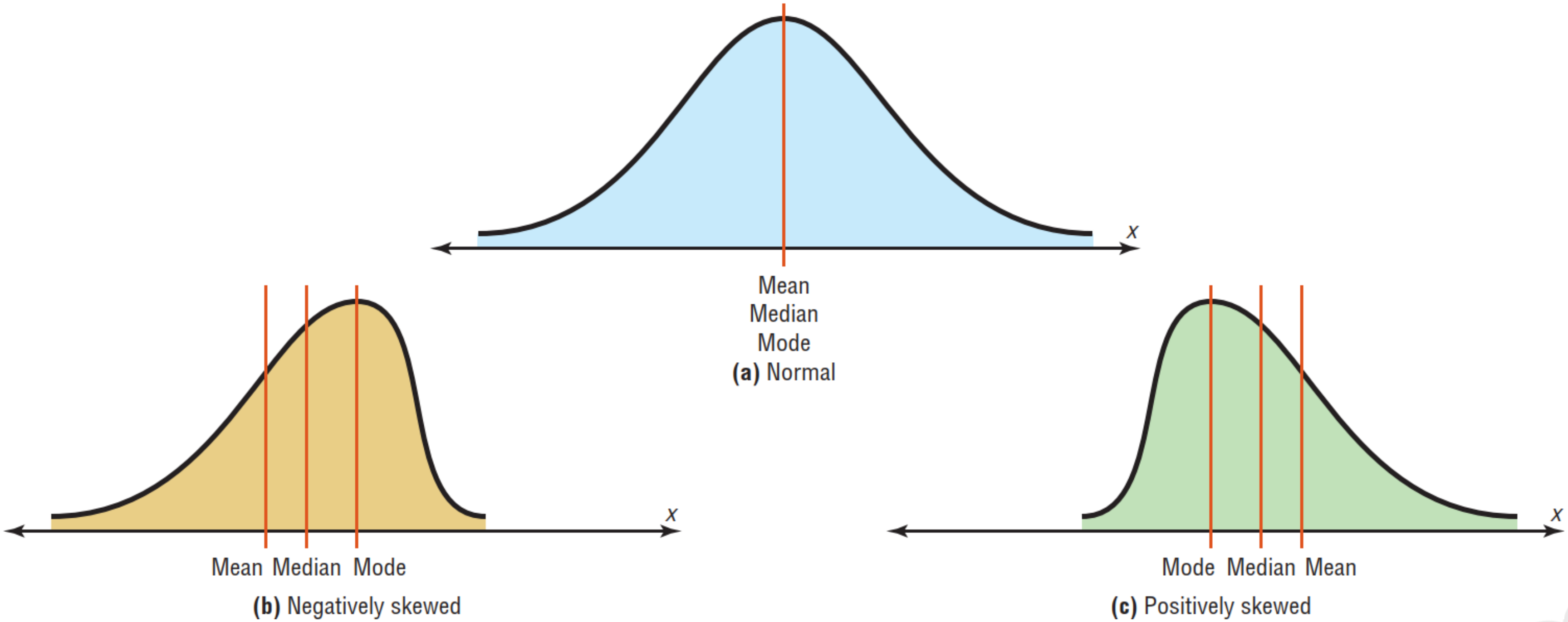
Summary of the Properties of the Theoretical Normal Distribution

1. A normal distribution curve is bell-shaped.
2. The mean, median, and mode are equal and are located at the center of the distribution.
3. A normal distribution curve is unimodal (i.e., it has only one mode).
4. The curve is symmetric about the mean, which is equivalent to saying that its shape is the same on both sides of a vertical line passing through the center.
5. The curve is continuous; that is, there are no gaps or holes. For each value of X, there is a corresponding value of Y.
6. The curve never touches the x axis. Theoretically, no matter how far in either direction the curve extends, it never meets the x axis—but it gets increasingly close.
7. The total area under a normal distribution curve is equal to 1.00, or 100%. This fact may seem unusual, since the curve never touches the x axis, but one can prove it mathematically by using calculus.
8. The area under the part of a normal curve that lies within 1 standard deviation of the mean is approximately 0.68, or 68%; within 2 standard deviations, about 0.95, or 95%; and within 3 standard deviations, about 0.997, or 99.7%.

Areas under a normal distribution curve, revisit lecture 2, slide 79

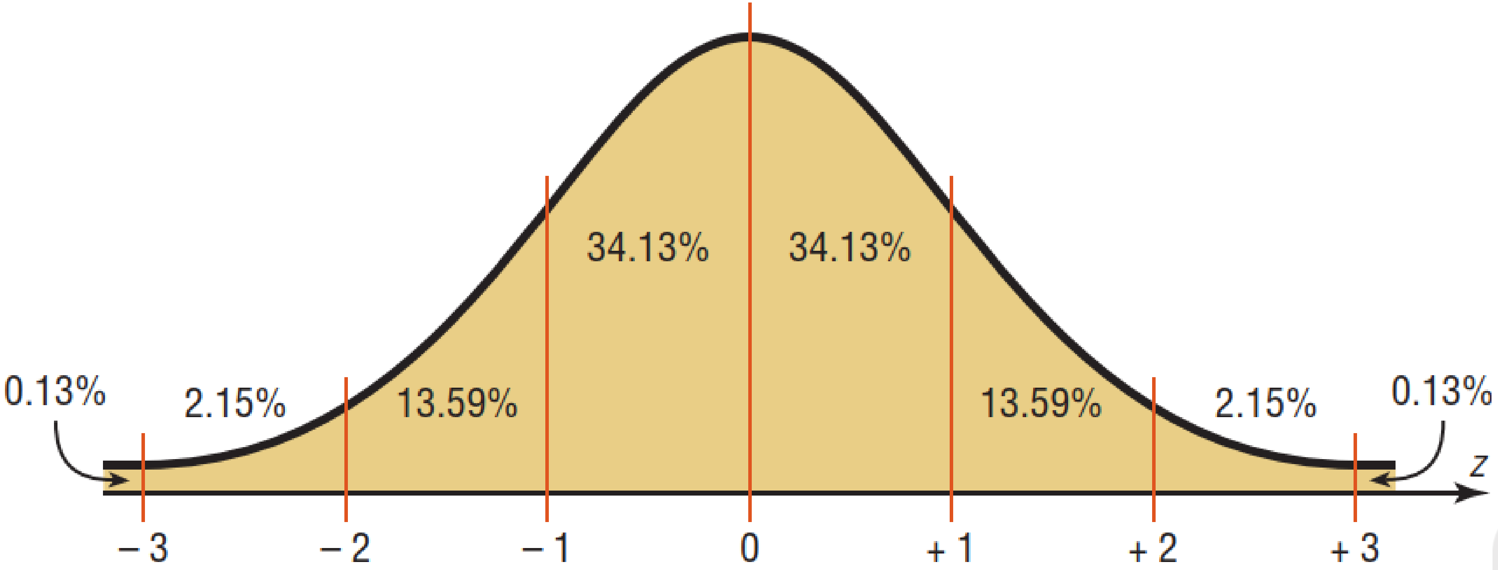


Normal and skewed distributions, revisit lecture 2, slides 43-45



The standard normal distribution

The **standard normal distribution** is a normal distribution with a mean of 0 and a standard deviation of 1.



The standard normal distribution

The formula for the standard normal distribution is

$$y = \frac{e^{-z^2/2}}{\sqrt{2\pi}}$$

All normally distributed variables can be transformed into the standard normally distributed variable by using the formula for the standard score:

$$z = \frac{\text{value} - \text{mean}}{\text{standard deviation}} \quad \text{or} \quad z = \frac{X - \mu}{\sigma}$$

The standard normal distribution

Finding Areas Under the Standard Normal Distribution Curve

For the solution of problems using the standard normal distribution, a two-step process is recommended with the use of the Procedure Table shown.

The two steps are as follows:

Step 1 Draw the normal distribution curve and shade the area.

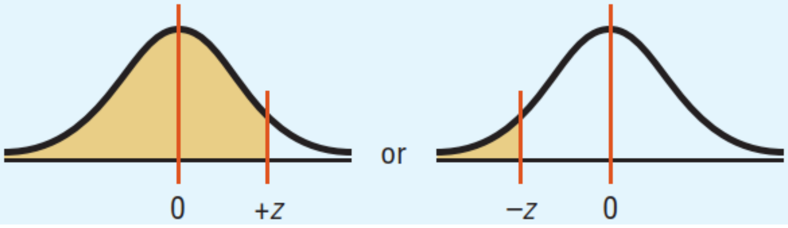
Step 2 Find the appropriate figure in the Procedure Table and follow the directions given.

The standard normal distribution

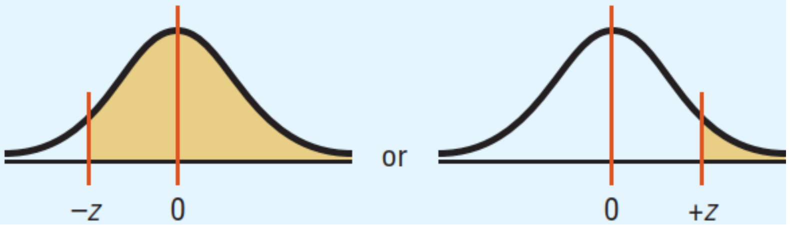
Procedure Table

Finding the Area Under the Standard Normal Distribution Curve

1. To the left of any z value:
   Look up the z value in the table and use the area given.



2. To the right of any z value:
   Look up the z value and subtract the area from 1.



3. Between any two z values:
   Look up both z values and subtract the corresponding areas.

Portions of the cumulative standard normal distribution

The Standard Normal Distribution

Cumulative Standard Normal Distribution

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
−3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
−3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
−3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
−3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
−2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
−2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
−2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
−2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
−2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
−2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
−2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
−2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110

Portions of the cumulative standard normal distribution

The Standard Normal Distribution (*continued*)

Cumulative Standard Normal Distribution

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
−2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
−1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
−1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
−1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
−1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
−1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
−1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
−1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
−1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
−1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
−1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379

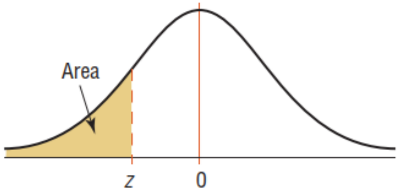
Portions of the cumulative standard normal distribution

The Standard Normal Distribution (*continued*)

Cumulative Standard Normal Distribution

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
−0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
−0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
−0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
−0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
−0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
−0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
−0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
−0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
−0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

For *z* values less than −3.49, use 0.0001.



Portions of the cumulative standard normal distribution

The Standard Normal Distribution *(continued)*

Cumulative Standard Normal Distribution

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015

Portions of the cumulative standard normal distribution

The Standard Normal Distribution ***(continued)***

Cumulative Standard Normal Distribution

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936

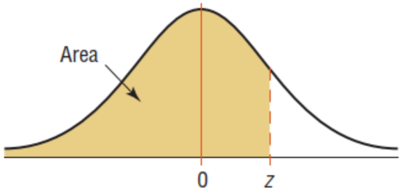
Portions of the cumulative standard normal distribution

The Standard Normal Distribution (*continued*)

Cumulative Standard Normal Distribution

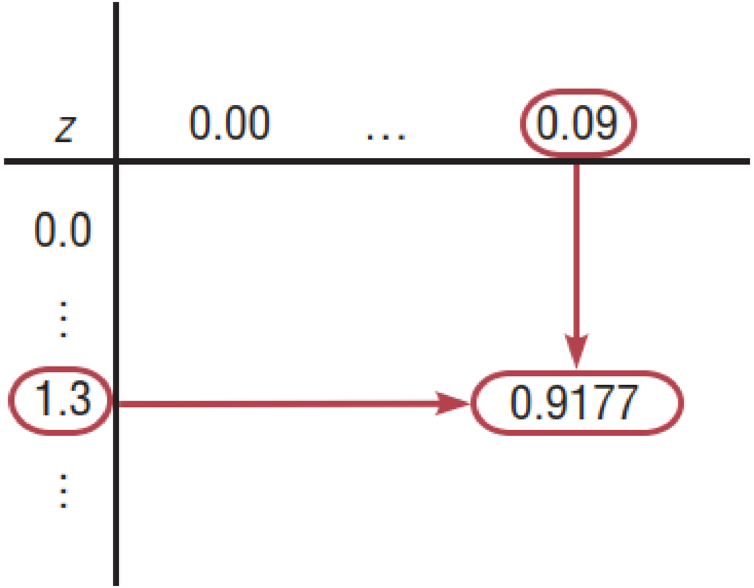
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

For *z* values greater than 3.49, use 0.9999.



Using the standard normal distribution table, revisit slide 19

For example, the area to the left of a z value of 1.39 is found by looking up 1.3 in the left column and 0.09 in the top row. Where the row and column lines meet gives an area of 0.9177.



Area Value for $z = 1.39$

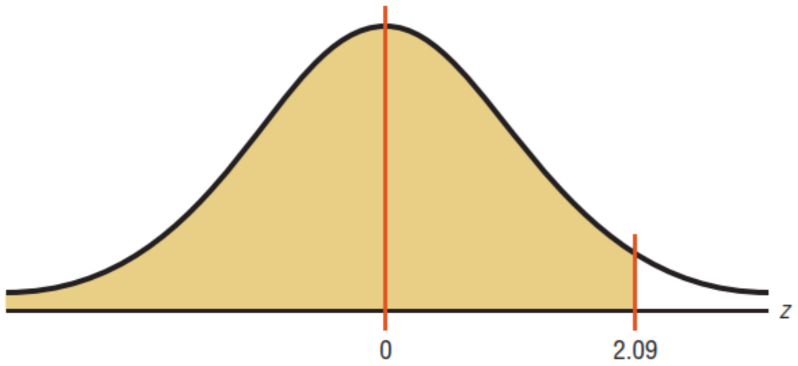
Example 1, revisit slide 19

Find the area under the standard normal distribution curve to the left of $z = 2.09$.

SOLUTION

Step 1 Draw the figure. The desired area is shown.

Area Under the Standard Normal Distribution Curve for Example 1



Step 2 We are looking for the area under the standard normal distribution curve to the left of $z = 2.09$. Since this is an example of the first case, look up the area in the table. It is 0.9817. Hence, 98.17% of the area is to the left of $z = 2.09$.

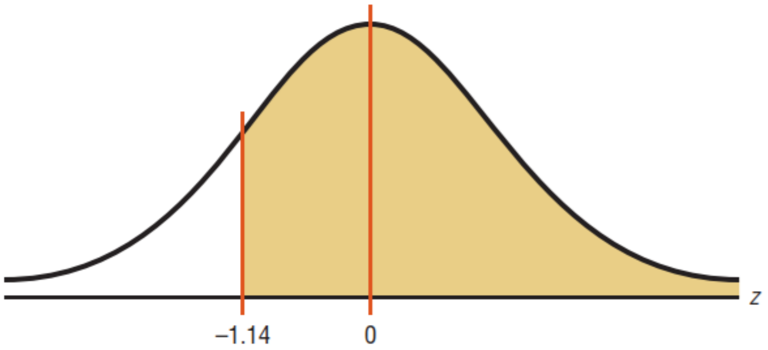
Example 2, revisit slide 16

Find the area under the standard normal distribution curve to the right of $z = -1.14$.

SOLUTION

Step 1 Draw the figure. The desired area is shown in Figure.

Area Under the Standard Normal Distribution Curve for Example 2



Step 2 We are looking for the area to the right of $z = -1.14$. This is an example of the second case. Look up the area for $z = -1.14$. It is 0.1271. Subtract it from 1.0000: $1.0000 - 0.1271 = 0.8729$. Hence, 87.29% of the area under the standard normal distribution curve is to the right of $z = -1.14$.

Note: In this situation, we subtract the area 0.1271 from 1.0000 because the table gives the area up to -1.14. So, to get the area under the curve to the right of -1.14, subtract the area 0.1271 from 1.0000, the total area under the standard normal distribution curve.

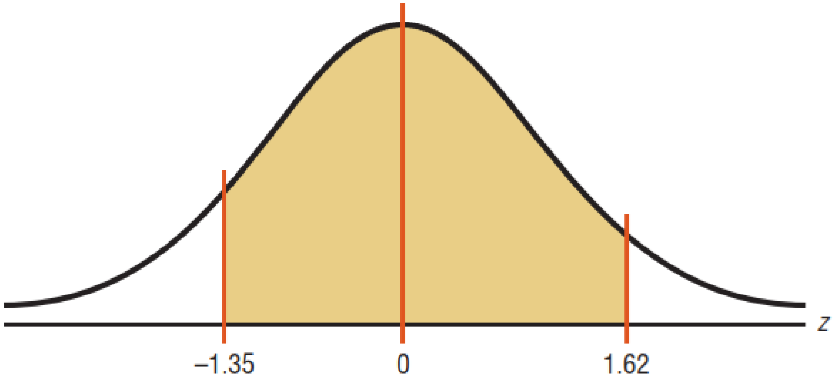
Example 3, revisit slides 16, 19

Find the area under the standard normal distribution curve between $z = 1.62$ and $z = -1.35$.

SOLUTION

Step 1 Draw the figure as shown. The desired area is shown in Figure.

Area Under the Standard Normal Distribution Curve for Example 3



Step 2 Since the area desired is between two given z values, look up the areas corresponding to the two z values and subtract the smaller area from the larger area. (Do not subtract the z values.) The area for $z = 1.62$ is 0.9474, and the area for $z = -1.35$ is 0.0885. The area between the two z values is $0.9474 - 0.0885 = 0.8589$, or 85.89%.

Example 4, revisit slides 19, 20

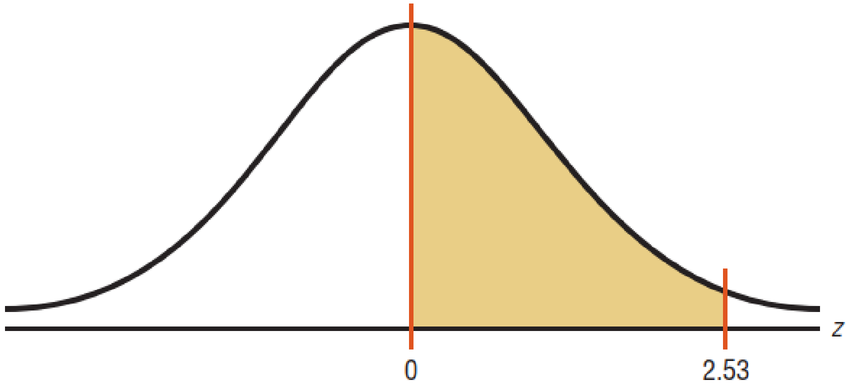
Find the probability for each. (Assume this is a standard normal distribution.)

a. $P(0 < z < 2.53)$ *b.* $P(z < 1.73)$ *c.* $P(z > 1.98)$

SOLUTION

a. $P(0 < z < 2.53)$ is used to find the area under the standard normal distribution curve between $z = 0$ and $z = 2.53$. First, draw the curve and shade the desired area. This is shown in Figure. Second, find the area in Table corresponding to $z = 2.53$. It is 0.9943. Third, find the area in Table corresponding to $z = 0$. It is 0.5000. Finally, subtract the two areas: $0.9943 - 0.5000 = 0.4943$. Hence, the probability is 0.4943, or 49.43%.

Area Under the Standard Normal Distribution Curve for Part *a* of Example 4

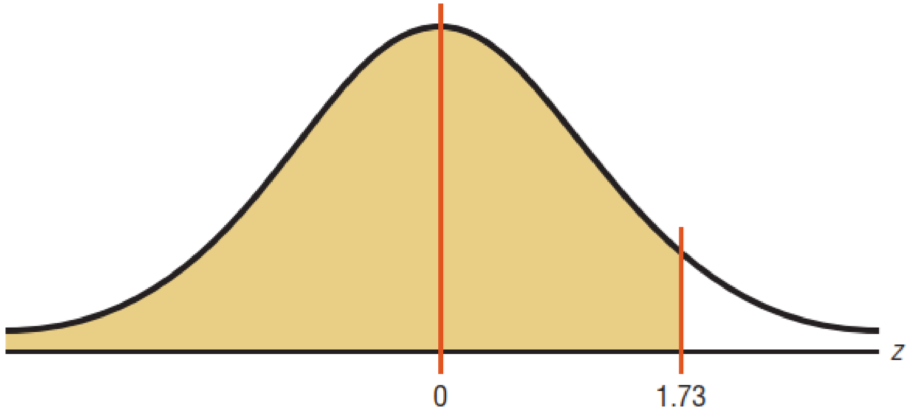


Example 4

SOLUTION

b. $P(z < 1.73)$ is used to find the area under the standard normal distribution curve to the left of $z = 1.73$. First, draw the curve and shade the desired area.
Second, find the area in Table corresponding to 1.73. It is 0.9582.
Hence, the probability of obtaining a z value less than 1.73 is 0.9582, or 95.82%.

Area Under the Standard Normal Distribution Curve for Part *b* of Example 4

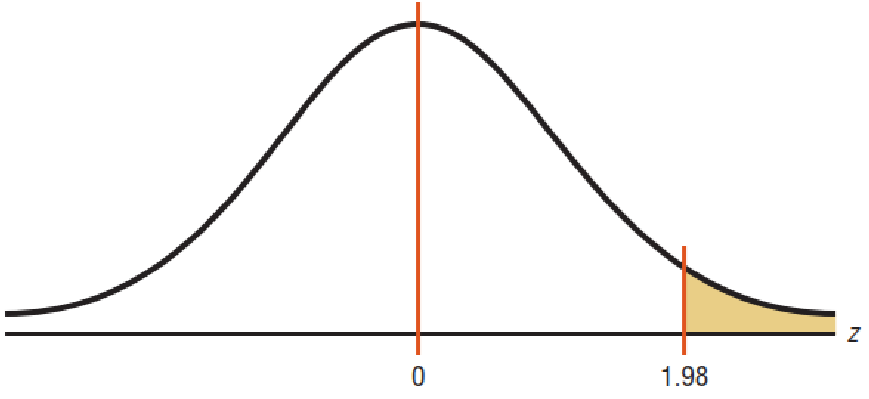


Example 4

SOLUTION

c. $P(z > 1.98)$ is used to find the area under the standard normal distribution curve to the right of $z = 1.98$. First, draw the curve and shade the desired area. Second, find the area corresponding to $z = 1.98$ in Table. It is 0.9761. Finally, subtract this area from 1.0000. It is $1.0000 - 0.9761 = 0.0239$. Hence, the probability of obtaining a z value greater than 1.98 is 0.0239, or 2.39%.

Area Under the Standard Normal Distribution Curve for Part *c* of Example 4



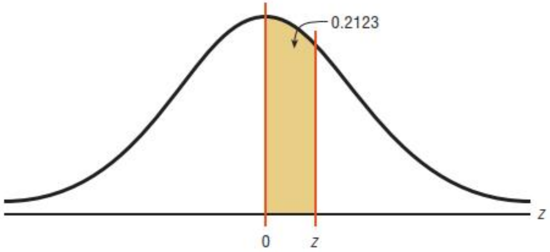
Example 5, revisit slide 18

Find the z value such that the area under the standard normal distribution curve between 0 and the z value is 0.2123.

SOLUTION

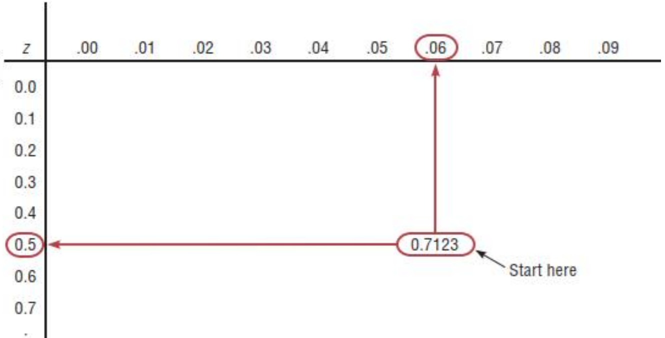
Draw the figure. The area is shown in Figure.

Area Under the Standard Normal Distribution Curve for Example 5



In this case it is necessary to add 0.5000 to the given area of 0.2123 to get the cumulative area of 0.7123. Look up the area in Table. The value in the left column is 0.5, and the top value is 0.06. Add these two values to get $z = 0.56$.

Finding the z Value from Table for Example 5



Standardization

Find probabilities for a normally distributed variable by transforming it into a standard normal variable.

To solve problems by using the standard normal distribution, transform the original variable to a standard normal distribution variable by using the formula

$$z = \frac{\text{value} - \text{mean}}{\text{standard deviation}} \quad \text{or} \quad z = \frac{X - \mu}{\sigma}$$

For example, suppose that the scores for a standardized test are normally distributed, have a mean of 100, and have a standard deviation of 15. When the scores are transformed to z values, the two distributions coincide, as shown in Figure. (Recall that the z distribution has a mean of 0 and a standard deviation of 1.)

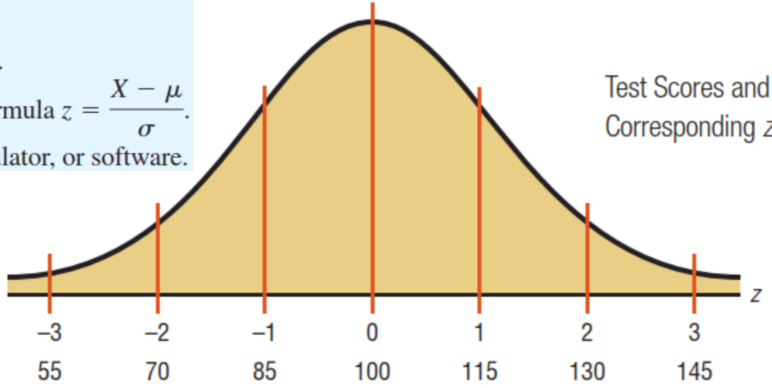
Procedure Table

Finding the Area Under Any Normal Curve

Step 1 Draw a normal curve and shade the desired area.

Step 2 Convert the values of X to z values, using the formula $z = \frac{X - \mu}{\sigma}$.

Step 3 Find the corresponding area, using a table, calculator, or software.



Test Scores and Their Corresponding z Values

Note: The z values are rounded to two decimal places.

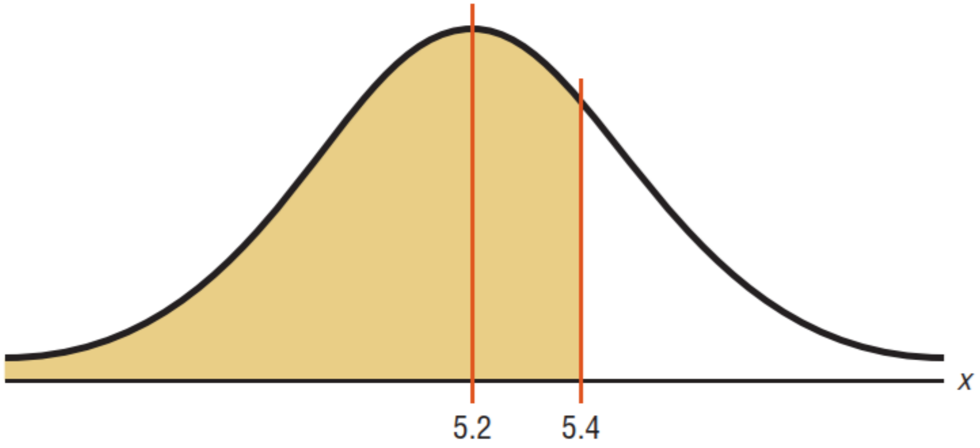
Example 6, revisit slides 18, 29

An adult has on average 5.2 liters of blood. Assume the variable is normally distributed and has a standard deviation of 0.3. Find the percentage of people who have less than 5.4 liters of blood in their system.

SOLUTION

Step 1 Draw a normal curve and shade the desired area.

Area Under a Normal Curve for Example 6



Example 6

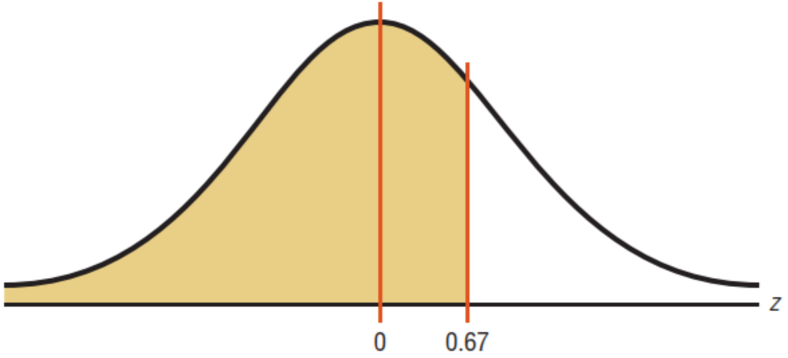
SOLUTION

Step 2 Find the z value corresponding to 5.4.

$$z = \frac{X - \mu}{\sigma} = \frac{5.4 - 5.2}{0.3} = \frac{0.2}{0.3} = 0.67$$

Hence, 5.4 is 0.67 of a standard deviation above the mean.

Area and z Values for Example 6



Step 3 Find the corresponding area in Table. The area under the standard normal curve to the left of $z = 0.67$ is 0.7486.

Therefore, 0.7486, or 74.86%, of adults have less than 5.4 liters of blood in their system.

Example 7, revisit slides 17, 18, 19, 29

Each month, an American household generates an average of 28 pounds of newspaper for garbage or recycling. Assume the variable is approximately normally distributed and the standard deviation is 2 pounds. If a household is selected at random, find the probability of its generating

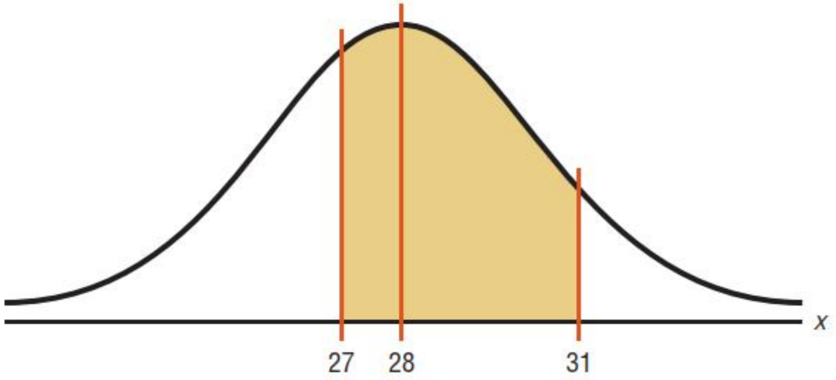
a. Between 27 and 31 pounds per month

b. More than 30.2 pounds per month

SOLUTION *a*

Step 1 Draw a normal curve and shade the desired area.

Area Under a Normal Curve for Part *a* of Example 7



Example 7

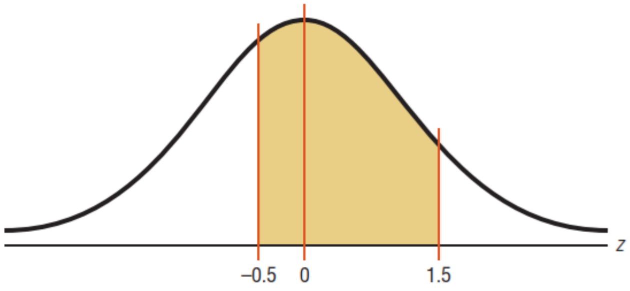
SOLUTION a **Step 2** Find the two z values.

$$z_1 = \frac{X - \mu}{\sigma} = \frac{27 - 28}{2} = -\frac{1}{2} = -0.5$$

$$z_2 = \frac{X - \mu}{\sigma} = \frac{31 - 28}{2} = \frac{3}{2} = 1.5$$

Step 3 Find the appropriate area, using Table. The area to the left of z_2 is 0.9332, and the area to the left of z_1 is 0.3085. Hence, the area between z_1 and z_2 is $0.9332 - 0.3085 = 0.6247$.

Area and z Values for Part *a* of Example 7



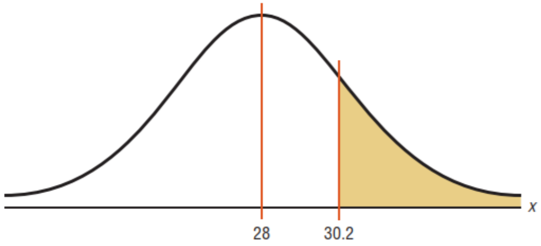
Hence, the probability that a randomly selected household generates between 27 and 31 pounds of newspapers per month is 62.47%.

Example 7

SOLUTION b

Step 1 Draw a normal curve and shade the desired area, as shown in Figure.

Area Under a Normal Curve for Part *b* of Example 7

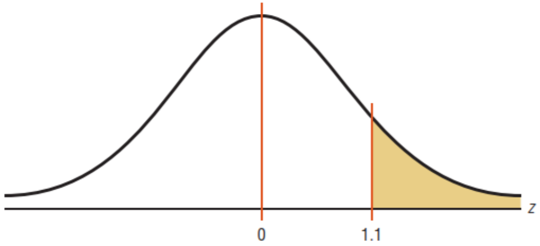


Step 2 Find the z value for 30.2.

$$z = \frac{X - \mu}{\sigma} = \frac{30.2 - 28}{2} = \frac{2.2}{2} = 1.1$$

Step 3 Find the appropriate area. The area to the left of $z = 1.1$ is 0.8643. Hence, the area to the right of $z = 1.1$ is $1.0000 - 0.8643 = 0.1357$.

Area and z Values for Part *b* of Example 7



Hence, the probability that a randomly selected household will accumulate more than 30.2 pounds of newspapers is 0.1357, or 13.57%.

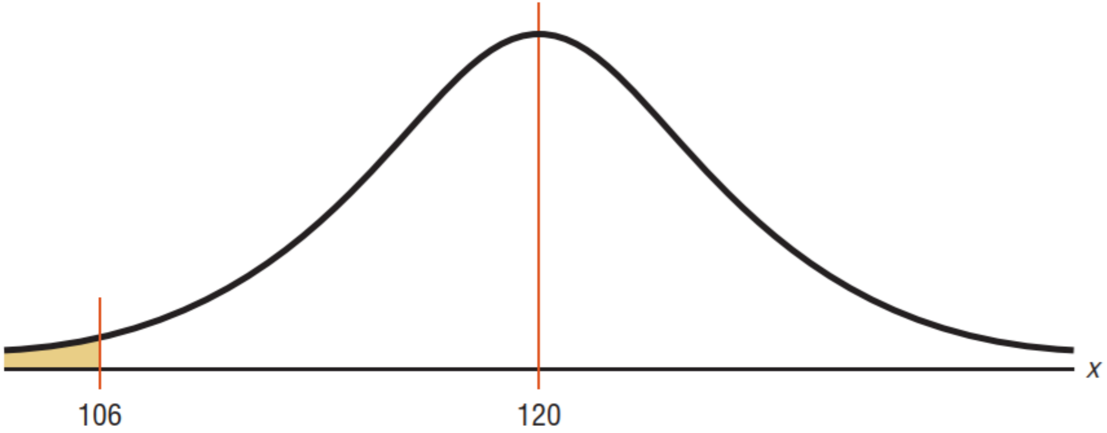
Example 8, revisit slides 15, 29

A desktop PC uses 120 watts of electricity per hour based on 4 hours of use per day. Assume the variable is approximately normally distributed and the standard deviation is 6 watts. If 500 PCs are selected, approximately how many will use less than 106 watts of power?

SOLUTION

Step 1 Draw a normal curve and shade the desired area.

Area Under a Normal Curve for Example 8



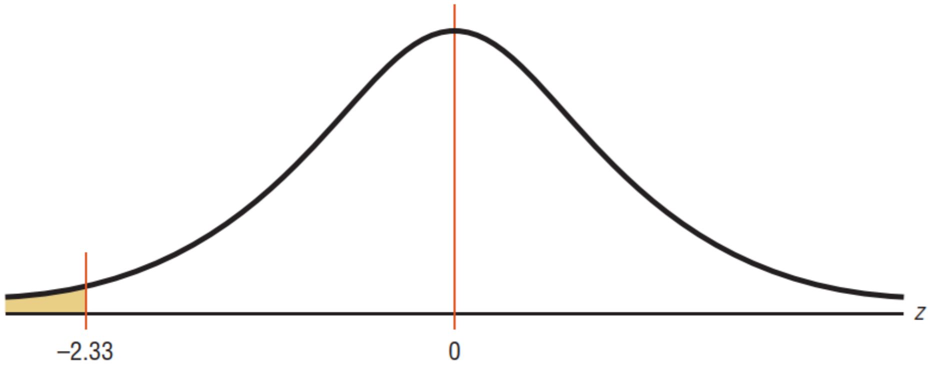
Example 8

SOLUTION **Step 2** Find the z value corresponding to 106.

$$z = \frac{X - \mu}{\sigma} = \frac{106 - 120}{6} = -2.33$$

Step 3 Find the area corresponding to $z = -2.33$. It is 0.0099.

Area and z Values for Example 8



To find the number of PCs that use less than 106 watts per hour, multiply $500 \times 0.0099 = 4.95$. Since we are using PCs, round the answer to 5. Hence, approximately 5 PCs use less than 106 watts.

Finding Data Values Given Specific Probabilities

A normal distribution can also be used to find specific data values for given percentages. In this case, you are given a probability or percentage and need to find the corresponding data value X. You can use the formula $z = \frac{X - \mu}{\sigma}$. Substitute the values for z, μ, and σ; then solve for X. However, you can transform the formula and solve for X as shown.

$$z = \frac{X - \mu}{\sigma}$$

$$z \cdot \sigma = X - \mu \qquad \text{Multiply both sides by } \sigma.$$

$$z \cdot \sigma + \mu = X - \mu + \mu \qquad \text{Add } \mu \text{ to both sides.}$$

$$X = z \cdot \sigma + \mu \qquad \text{Exchange both sides of the equation.}$$

Now you can use this new formula and find the value for X.

Finding data values (given percentages) via the standard normal distribution

Formula for Finding the Value of a Normal Variable *X*

$$X = z \cdot \sigma + \mu$$

The complete procedure for finding an X value is summarized in the Procedure Table shown.

Procedure Table

Finding Data Values for Specific Probabilities

Step 1 Draw a normal curve and shade the desired area that represents the probability, proportion, or percentile.

Step 2 Find the z value from the table that corresponds to the desired area.

Step 3 Calculate the X value by using the formula $X = z\sigma + \mu$.

Example 9, revisit slides 18, 37, 38

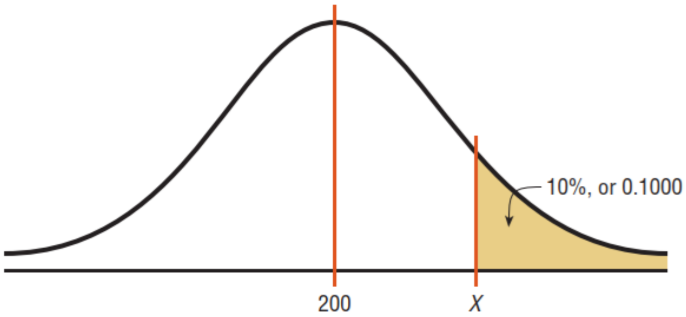
To qualify for a police academy, candidates must score in the top 10% on a general abilities test. Assume the test scores are normally distributed and the test has a mean of 200 and a standard deviation of 20. Find the lowest possible score to qualify.

SOLUTION

Step 1 Draw a normal distribution curve and shade the desired area that represents the probability.

Since the test scores are normally distributed, the test value X that cuts off the upper 10% of the area under a normal distribution curve is desired. This area is shown.

Area Under a Normal Curve for Example 9



Work backward to solve this problem.

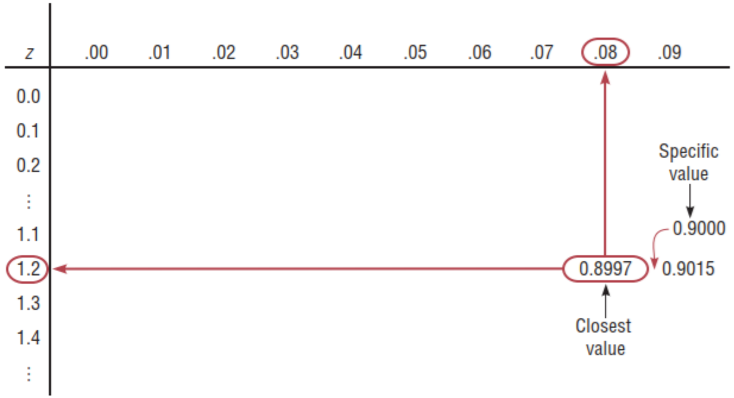
Subtract 0.1000 from 1.0000 to get the area under the normal distribution to the left of x: $1.0000 - 0.1000 = 0.9000$.

Example 9

SOLUTION **Step 2** Find the z value from Table that corresponds to the desired area.

Find the z value that corresponds to an area of 0.9000 by looking up 0.9000 in the area portion of Table. If the specific value cannot be found, use the closest value—in this case 0.8997, as shown. The corresponding z value is 1.28. (If the area falls exactly halfway between two z values, use the larger of the two z values. For example, the area 0.9500 falls halfway between 0.9495 and 0.9505. In this case use 1.65 rather than 1.64 for the z value.)

Finding the z Value from



Step 3 Find the X value.

$$X = z \cdot \sigma + \mu = 1.28(20) + 200 = 25.6 + 200$$
$$= 225.6 = 226 \text{ (rounded)}$$

A score of 226 should be used as a cutoff. Anybody scoring 226 or higher qualifies for the academy.

Example 10, revisit slides 17, 18, 37, 38

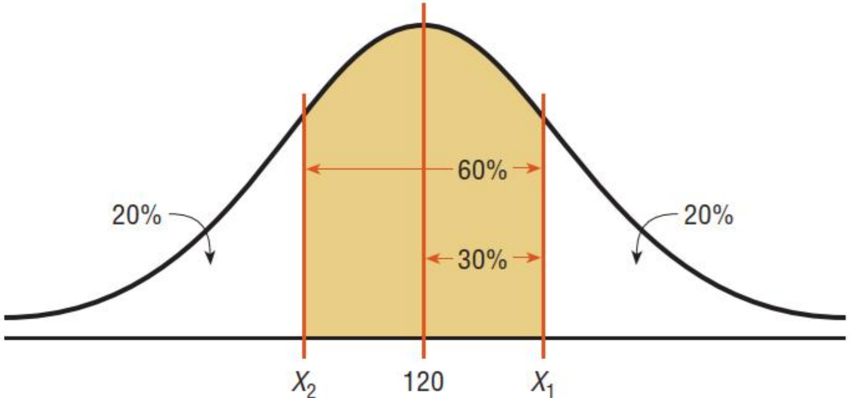
For a medical study, a researcher wishes to select people in the middle 60% of the population based on blood pressure. Assuming that blood pressure readings are normally distributed and the mean systolic blood pressure is 120 and the standard deviation is 8, find the upper and lower readings that would qualify people to participate in the study.

SOLUTION

Step 1 Draw a normal distribution curve and shade the desired area. The cutoff points are shown.

Two values are needed, one above the mean and one below the mean.

Area Under a Normal Curve for Example 10



Example 10

Step 2 Find the z values.

To get the area to the left of the positive z value, add $0.5000 + 0.3000 = 0.8000$ ($30\% = 0.3000$). The z value with area to the left closest to 0.8000 is 0.84.

Step 3 Calculate the X values.

Substituting in the formula $X = z\sigma + \mu$ gives

$$X_1 = z\sigma + \mu = (0.84)(8) + 120 = 126.72$$

The area to the left of the negative z value is 20%, or 0.2000. The area closest to 0.2000 is -0.84.

$$X_2 = (-0.84)(8) + 120 = 113.28$$

Therefore, the middle 60% will have blood pressure readings of $113.28 < X < 126.72$.

The Central Limit Theorem

Sample means distribution

Distribution of Sample Means

Suppose a researcher selects a sample of 30 adult males and finds the mean of the measure of the triglyceride levels for the sample subjects to be 187 milligrams/deciliter. Then suppose a second sample is selected, and the mean of that sample is found to be 192 milligrams/deciliter. Continue the process for 100 samples. What happens then is that the mean becomes a random variable, and the sample means 187, 192, 184, . . . , 196 constitute a *sampling distribution of sample means.*

A **sampling distribution of sample means** is a distribution using the means computed from all possible random samples of a specific size taken from a population.

If the samples are randomly selected with replacement, the sample means, for the most part, will be somewhat different from the population mean μ. These differences are caused by sampling error.

Sampling error is the difference between the sample measure and the corresponding population measure due to the fact that the sample is not a perfect representation of the population.

Sample means distribution

When all possible samples of a specific size are selected with replacement from a population, the distribution of the sample means for a variable has two important properties, which are explained next.

Properties of the Distribution of Sample Means

1. The mean of the sample means will be the same as the population mean.
2. The standard deviation of the sample means will be smaller than the standard deviation of the population, and it will be equal to the population standard deviation divided by the square root of the sample size.

Sample means distribution

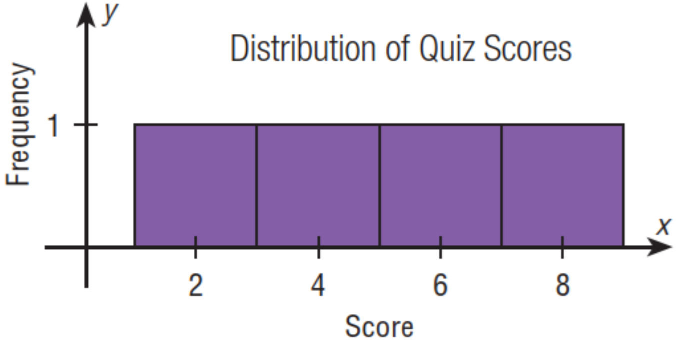
The following example illustrates these two properties. Suppose a professor gave an 8-point quiz to a small class of four students. The results of the quiz were 2, 6, 4, and 8. For the sake of discussion, assume that the four students constitute the population. The mean of the population is

$$\mu = \frac{2 + 6 + 4 + 8}{4} = 5$$

The standard deviation of the population is

$$\sigma = \sqrt{\frac{(2-5)^2 + (6-5)^2 + (4-5)^2 + (8-5)^2}{4}} \approx 2.236$$

The graph of the original distribution is shown. This is called a *uniform distribution.*



Sample means distribution

Now, if all samples of size 2 are taken with replacement and the mean of each sample is found, the distribution is as shown.

Sample	Mean	Sample	Mean
2, 2	2	6, 2	4
2, 4	3	6, 4	5
2, 6	4	6, 6	6
2, 8	5	6, 8	7
4, 2	3	8, 2	5
4, 4	4	8, 4	6
4, 6	5	8, 6	7
4, 8	6	8, 8	8

A frequency distribution of sample means is as follows.

$\bar{X}$	f
2	1
3	2
4	3
5	4
6	3
7	2
8	1

Sample means distribution

For the data from the example just discussed, the Figure shows the graph of the sample means. The histogram appears to be approximately normal.

The mean of the sample means, denoted by $\mu_{\overline{X}}$, is

$$\mu_{\overline{X}} = \frac{2 + 3 + \cdots + 8}{16} = \frac{80}{16} = 5$$

which is the same as the population mean. Hence,

$$\mu_{\overline{X}} = \mu$$

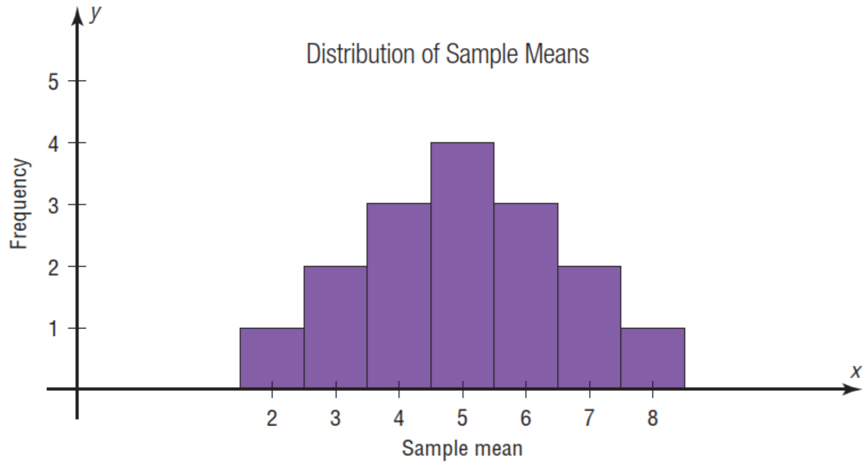
The standard deviation of sample means, denoted by $\sigma_{\overline{X}}$, is

$$\sigma_{\overline{X}} = \sqrt{\frac{(2-5)^2 + (3-5)^2 + \cdots + (8-5)^2}{16}} \approx 1.581$$

which is the same as the population standard deviation, divided by $\sqrt{2}$:

$$\sigma_{\overline{X}} = \frac{2.236}{\sqrt{2}} \approx 1.581$$

(*Note:* Rounding rules were not used here in order to show that the answers coincide.)



Sample means distribution

In summary, if all possible samples of size n are taken with replacement from the same population, the mean of the sample means, denoted by $\mu_{\overline{X}}$, equals the population mean μ; and the standard deviation of the sample means, denoted by $\sigma_{\overline{X}}$, equals $\sigma/\sqrt{n}$. The standard deviation of the sample means is called the **standard error of the mean.** Hence,

$$\sigma_{\overline{X}} = \frac{\sigma}{\sqrt{n}}$$

A third property of the sampling distribution of sample means pertains to the shape of the distribution and is explained by the **central limit theorem.**

The central limit theorem and applications

The Central Limit Theorem

As the sample size n increases without limit, the shape of the distribution of the sample means taken with replacement from a population with mean μ and standard deviation σ will approach a normal distribution. As previously shown, this distribution will have a mean μ and a standard deviation $\sigma/\sqrt{n}$.

If the sample size is sufficiently large, the central limit theorem can be used to answer questions about sample means in the same manner that a normal distribution can be used to answer questions about individual values. The only difference is that a new formula must be used for the z values. It is

$$z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}}$$

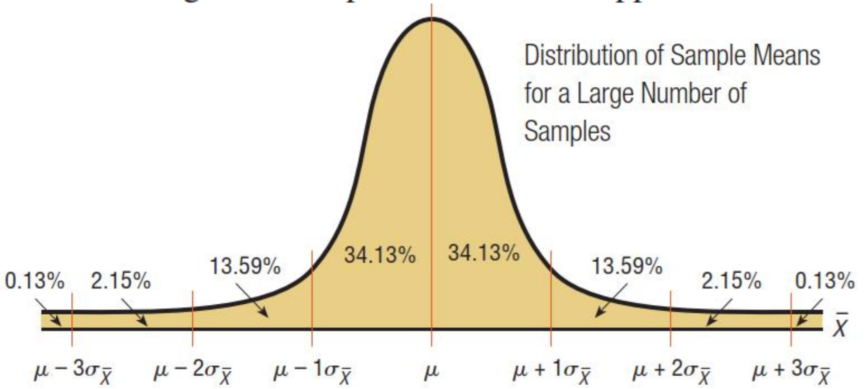
Notice that $\overline{X}$ is the sample mean, and the denominator must be adjusted since means are being used instead of individual data values. The denominator is the standard deviation of the sample means.

The central limit theorem and applications

If a large number of samples of a given size are selected from a normally distributed population, or if a large number of samples of a given size that is greater than or equal to 30 are selected from a population that is not normally distributed, and the sample means are computed, then the distribution of sample means will look like the one shown in Figure. Their percentages indicate the areas of the regions.

It's important to remember two things when you use the central limit theorem:

1. When the original variable is normally distributed, the distribution of the sample means will be normally distributed, for any sample size *n*.
2. When the distribution of the original variable is not normal, a sample size of 30 or more is needed to use a normal distribution to approximate the distribution of the sample means. The larger the sample, the better the approximation will be.



Example 11, revisit slides 19, 49

A. C. Neilsen reported that children between the ages of 2 and 5 watch an average of 25 hours of television per week. Assume the variable is normally distributed and the standard deviation is 3 hours. If 20 children between the ages of 2 and 5 are randomly selected, find the probability that the mean of the number of hours they watch television will be greater than 26.3 hours.

SOLUTION

Since the variable is approximately normally distributed, the distribution of sample means will be approximately normal, with a mean of 25. The standard deviation of the sample means is

$$\sigma_{\overline{X}} = \frac{\sigma}{\sqrt{n}} = \frac{3}{\sqrt{20}} = 0.671$$

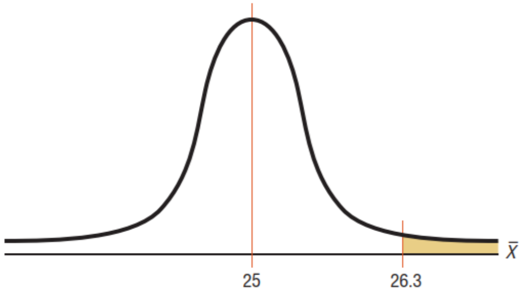
Example 11

SOLUTION

Step 1 Draw a normal curve and shade the desired area.

The distribution of the means is shown in Figure, with the appropriate area shaded.

Distribution of the Means for Example 11



Step 2 Convert the $\overline{X}$ value to a z value.

The z value is

$$z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} = \frac{26.3 - 25}{3/\sqrt{20}} = \frac{1.3}{0.671} = 1.94$$

Step 3 Find the corresponding area for the z value.

The area to the right of 1.94 is $1.000 - 0.9738 = 0.0262$, or 2.62%.

One can conclude that the probability of obtaining a sample mean larger than 26.3 hours is 2.62% [that is, $P(\overline{X} > 26.3) = 0.0262$]. Specifically, the probability that the 20 children selected between the ages of 2 and 5 watch more than 26.3 hours of television per week is 2.62%.

Example 12, revisit slides 15, 19, 49

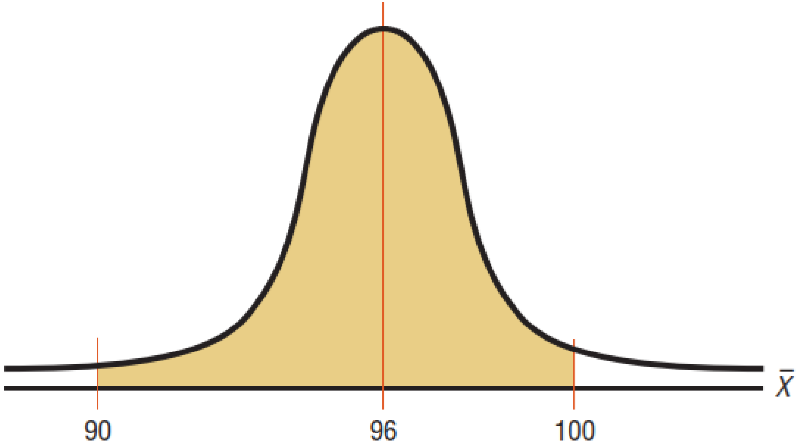
The average age of a vehicle registered in the United States is 8 years, or 96 months. Assume the standard deviation is 16 months. If a random sample of 36 vehicles is selected, find the probability that the mean of their age is between 90 and 100 months.

SOLUTION

Step 1 Draw a normal curve and shade the desired area.

Since the sample is 30 or larger, the normality assumption is not necessary. The desired area is shown.

Area Under a Normal Curve for Example 12



Example 12

SOLUTION **Step 2** Convert the $\overline{X}$ values to z values.

The two z values are

$$z_1 = \frac{90 - 96}{16/\sqrt{36}} = -2.25$$

$$z_2 = \frac{100 - 96}{16/\sqrt{36}} = 1.50$$

Step 3 Find the corresponding area for the z values.

To find the area between the two z values of -2.25 and 1.50, look up the corresponding areas in and subtract the smaller value from the larger value. The area for $z = -2.25$ is 0.0122, and the area for $z = 1.50$ is 0.9332. Hence, the area between the two values is $0.9332 - 0.0122 = 0.9210$, or 92.1%.

Hence, the probability of obtaining a sample mean between 90 and 100 months is 92.1%; that is, $P(90 < \overline{X} < 100) = 0.9210$. Specifically, the probability that the 36 vehicles selected have a mean age between 90 and 100 months is 92.1%.

Example 13, revisit slides 18, 49

The average time spent by construction workers who work on weekends is 7.93 hours (over 2 days). Assume the distribution is approximately normal and has a standard deviation of 0.8 hour.

a. Find the probability that an individual who works at that trade works fewer than 8 hours on the weekend.

b. If a sample of 40 construction workers is randomly selected, find the probability that the mean of the sample will be less than 8 hours.

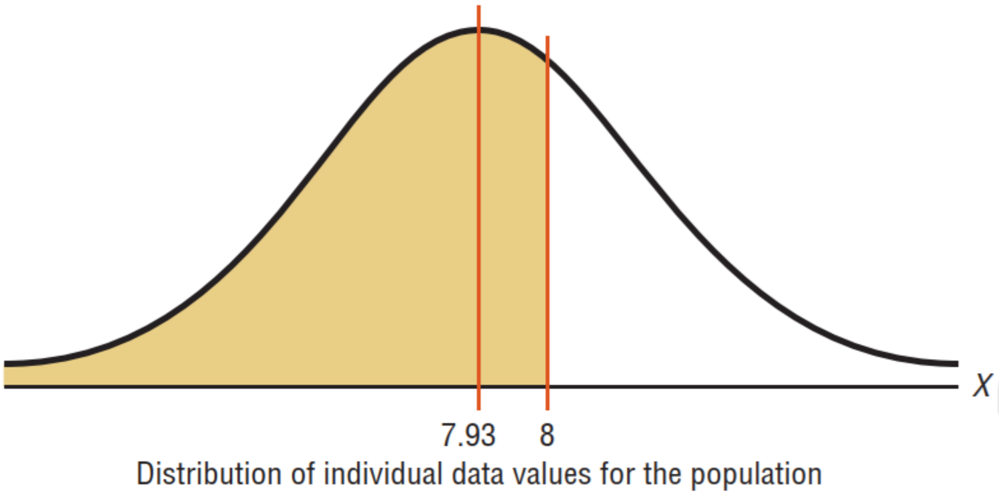
Example 13

SOLUTION *a*

Step 1 Draw a normal distribution and shade the desired area.

Since the question concerns an individual person, the formula $z = (X - \mu)/\sigma$ is used. The distribution is shown.

Area Under a Normal Curve for Part *a* of Example 13



Example 13

SOLUTION a

Step 2 Find the z value.

$$Z = \frac{X - \mu}{\sigma} = \frac{8 - 7.93}{0.8} \approx 0.09$$

Step 3 Find the area to the left of $z = 0.09$.

It is 0.5359.

Hence, the probability of selecting a construction worker who works less than 8 hours on a weekend is 0.5359, or 53.59%.

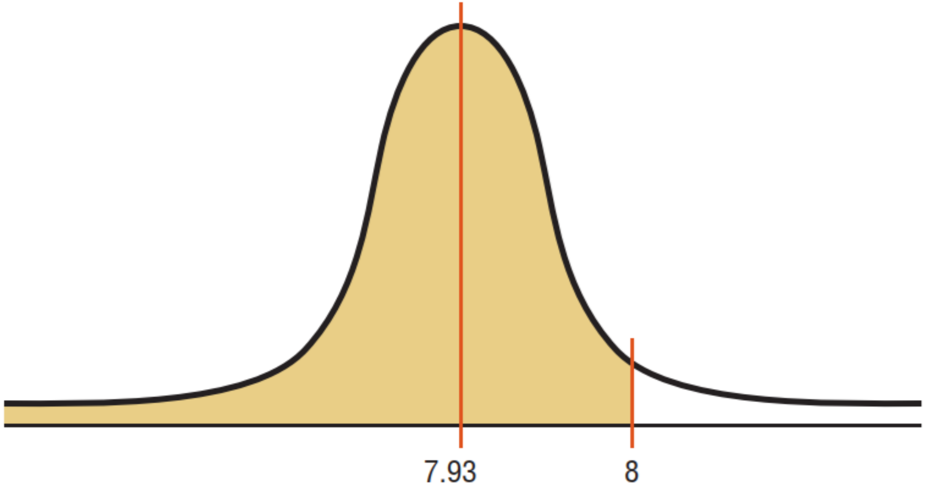
Example 13

SOLUTION *b*

Step 1 Draw a normal curve and shade the desired area.

Since the question concerns the mean of a sample with a size of 40, the central limit theorem formula $z = (\overline{X} - \mu)/(\sigma/\sqrt{n})$ is used. The area is shown in Figure.

Area Under a Normal Curve for Part *b* of Example 13



Distribution of means for all samples of size 40 taken from the population

Example 13

SOLUTION *b*

Step 2 Find the z value for a mean of 8 hours and a sample size of 40.

$$z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} = \frac{8 - 7.93}{0.8/\sqrt{40}} \approx 0.55$$

Step 3 Find the area corresponding to $z = 0.55$. The area is 0.7088.

Hence, the probability of getting a sample mean of less than 8 hours when the sample size is 40 is 0.7088, or 70.88%.

Comparing the two probabilities, you can see the probability of selecting an individual construction worker who works less than 8 hours on a weekend is 53.59%. The probability of selecting a random sample of 40 construction workers with a mean of less than 8 hours per week is 70.88%. This difference of 17.29% is due to the fact that the distribution of sample means is much less variable than the distribution of individual data values. The reason is that as the sample size increases, the standard deviation of the means decreases.